

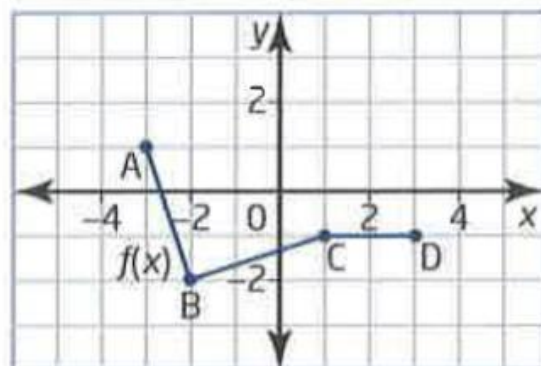
L2 - Transformations of Quadratic Functions – Lesson

MCR3U

DO IT NOW!

a) Complete the table of values for the function $f(x)$ and $g(x)$. Then use the table of values to plot image points and graph the function $g(x)$

$f(x) : (x, f(x))$	$g(x) : (x, f(x) + 4)$
A(-3, 1)	A'(-3, 5)



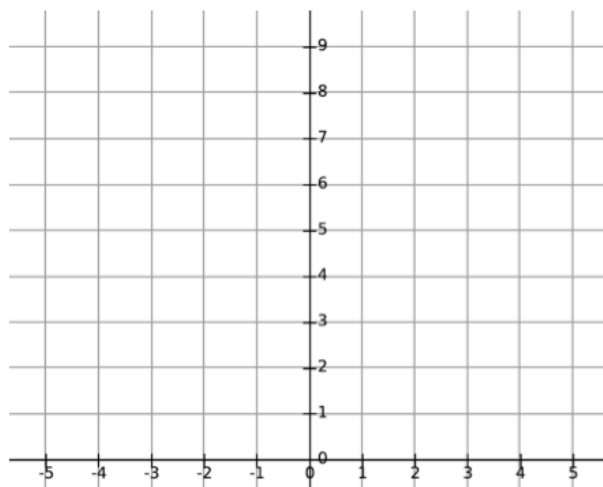
Quadratic Functions

Base Function:

Key Points:

x	y

Graph of Base Function:



Order of Transformations

1. stretches, compressions, reflections
2. translations

Example 1: If $f(x) = x^2$, describe the changes and write the transformed function:

a) $g(x) = 2f(x)$

b) $g(x) = f(2x)$

c) $g(x) = f(x) + 4$

d) $g(x) = f(x + 3)$

e) $g(x) = -f(x)$

f) $g(x) = f(-x)$

Example 2: For each of the following functions, describe the transformations to $f(x) = x^2$ in order and write the transformed equation.

a) $g(x) = -2f[-3(x + 3)] - 1$

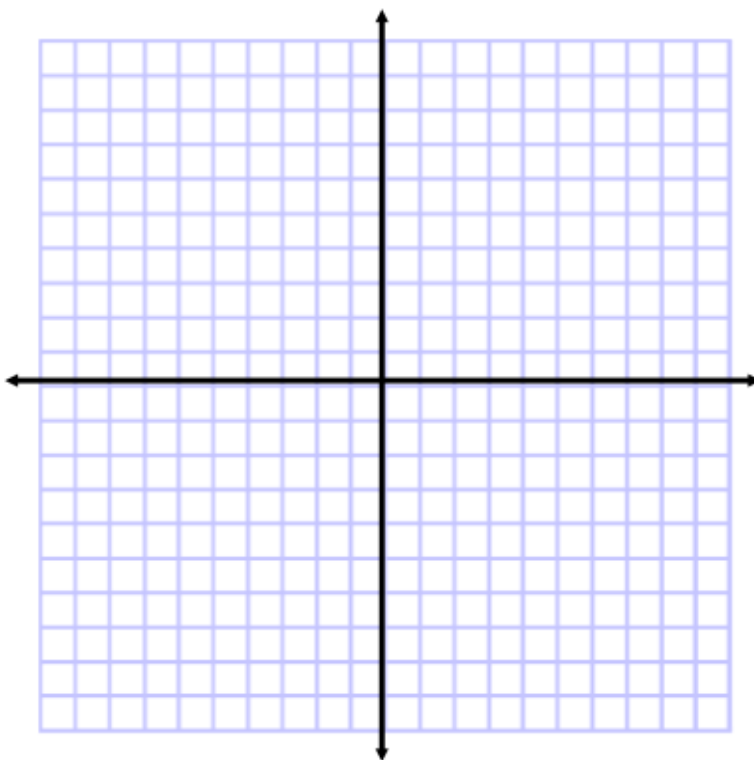
b) $y = \frac{1}{2}f[-3(x - 2)] + 5$

Example 3: for each of the following functions...

- i) make a table of values for the parent function
- ii) graph the parent function $f(x) = x^2$
- iii) describe the transformations
- iv) make a table of values of image points
- v) graph the transformed function and write it's equation

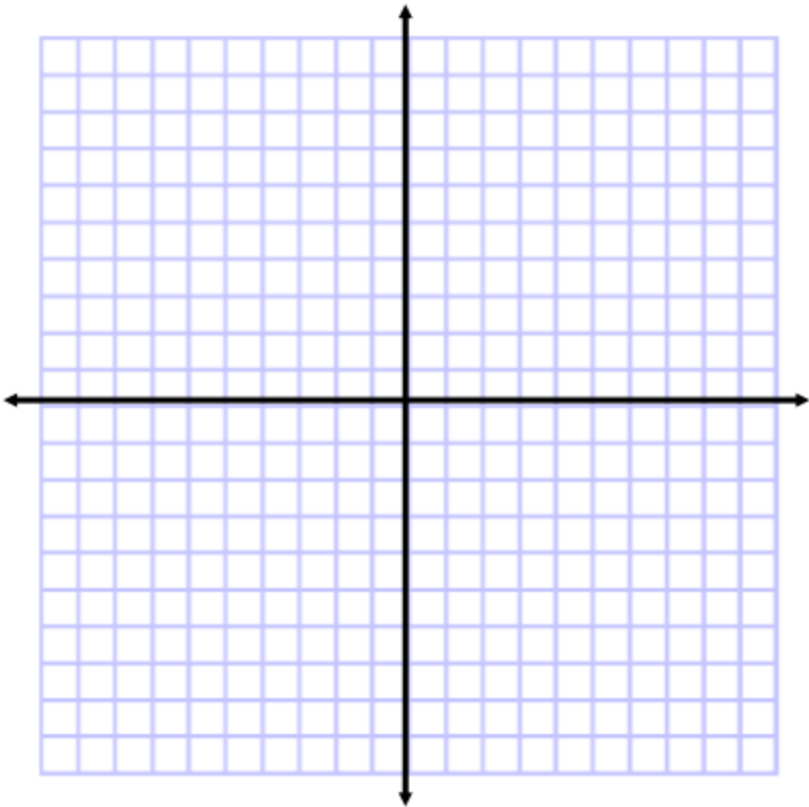
a) $g(x) = -f(2x)$

x	y



b) $g(x) = f\left[-\frac{1}{2}(x - 1)\right]$

<i>x</i>	<i>y</i>



c) $g(x) = -2f[-3(x + 3)] - 1$

<i>x</i>	<i>y</i>

